

## 7. Integration Techniques

### 7.3. Trigonometric Substitutions

#### 7.3a Learning Objectives

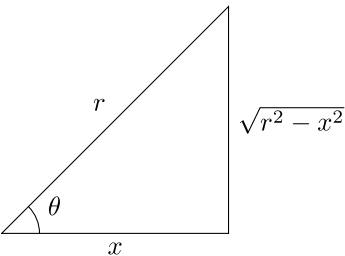
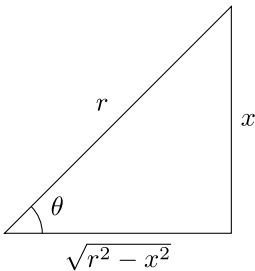
- I can understand the relationship between right triangles and substitutions.
- I can draw triangles to represent substitutions.
- I can understand when and how to use trigonometric substitutions.

Suppose we want to derive the formula for area of a circle of radius  $r$ . We know how to set up the integral for this using area between curves:

$$A = \int_{-r}^r \underbrace{\sqrt{r^2 - x^2}}_{\text{top}} - \underbrace{(-\sqrt{r^2 - x^2})}_{\text{bottom}} dx$$

$$= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx$$

Up until now, we don't have a method to evaluate this integral. Looking at a point on the circle we can use trig functions to rewrite the expressions in  $x$  and  $y$  into  $r$  and  $\theta$ . We have a few options, so let's do both and compare.

| Step                                 | Cosine                                                                                            | Sine                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| substitution                         | $x = r \cos(\theta)$                                                                              | $x = r \sin(\theta)$                                                                                                         |
| triangle                             |                |                                         |
| differential                         | $dx = -r \sin(\theta) d\theta$                                                                    | $dx = r \cos(\theta) d\theta$                                                                                                |
| bounds for $\theta$                  | $x = -r \Rightarrow \theta = \pi$<br>$x = r \Rightarrow \theta = 0$                               | $x = -r \Rightarrow \theta = 0$<br>$x = r \Rightarrow \theta = \frac{\pi}{2}$                                                |
| integral after trig sub steps below! | $2 \int_{\pi}^0 r \sin(\theta) (-r \sin(\theta)) d\theta$                                         | $2 \int_0^{\frac{\pi}{2}} r \cos(\theta) r \cos(\theta) d\theta$                                                             |
| integral after trig identity         | $2r^2(-1) \int_{\pi}^0 \frac{1}{2}(1 - \cos(2\theta)) d\theta$                                    | $2r^2 \int_0^{\frac{\pi}{2}} \frac{1}{2}(1 + \cos(2\theta)) d\theta$                                                         |
| Observations                         | <ul style="list-style-type: none"> <li>• bounds are flipped</li> <li>• extra negatives</li> </ul> | <ul style="list-style-type: none"> <li>• the bounds are least to greatest</li> <li>• there are no extra negatives</li> </ul> |

For  $x = r \sin(\theta)$  we get  $dx = r \cos(\theta) d\theta$ . We have to solve little equations for  $\theta$  to get our new bounds. Then the integral becomes

$$\begin{aligned}
 A &= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx \\
 &= 2 \int_0^\pi \sqrt{r^2 - (r \sin(\theta))^2} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^\pi \sqrt{r^2 - r^2 \sin^2(\theta)} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^\pi \sqrt{r^2(1 - \sin^2(\theta))} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^\pi \sqrt{r^2 \cos^2(\theta)} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^\pi r |\cos(\theta)| (r \cos(\theta)) d\theta \\
 &= 2 \int_0^\pi r^2 \cos^2(\theta) d\theta \\
 &= 2 \int_0^\pi r^2 \left[ \frac{1}{2}(1 + \cos(2\theta)) \right] d\theta \\
 &= r^2 \int_0^\pi (1 + \cos(2\theta)) d\theta
 \end{aligned}$$

Then we can finish this with a quick  $u$ -substitution, where  $u = 2\theta$  and  $du = 2 d\theta$  and bounds become  $u(0) = 0$  and  $u(\pi) = 2\pi$ . Then

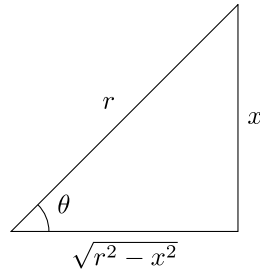
$$\begin{aligned}
 A &= \frac{1}{2} r^2 \int_0^{2\pi} 1 + \cos(u) du \\
 &= \frac{1}{2} r^2 (u + \sin(u))_0^{2\pi} \\
 &= \frac{1}{2} r^2 (2\pi + 0 - 0 + 0) \\
 &= \pi r^2
 \end{aligned}$$

And so we can get the area of the circle! You can try doing the other substitution if you want, using the trigonometric identity  $\frac{1}{2}(1 - \cos(2\theta)) = \sin^2(\theta)$ .

Ultimately, either substitution **could** work **if you are careful enough**. However, we will just show the substitutions and patterns for the trig subs that don't give extra minus signs. If you are still confused on why we do things in the order we do them, please come chat in office hours about it! (It honestly took me a while to figure this out myself!)

Here is a “more elegant” way to solve this problem, by which we mean a way that takes a lot fewer steps and uses the triangle a bit more:

We set up the integral the same way, but using trigonometric relations and the triangle reproduced here



We can write

$$\cos(\theta) = \frac{\sqrt{r^2 - x^2}}{r} \Rightarrow \sqrt{r^2 - x^2} = r \cos(\theta)$$

Combining this with the differential,  $dx = r \cos(\theta) d\theta$ , from before, we can jump straight to

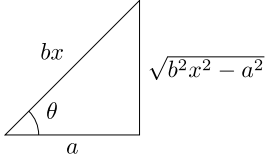
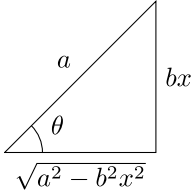
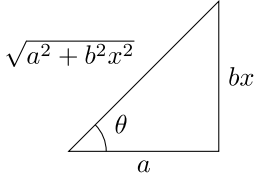
$$\begin{aligned} A &= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx \\ &= 2 \int_0^\pi r \cos(\theta) r \cos(\theta) d\theta \\ &= 2r^2 \int_0^\pi \cos^2(\theta) d\theta \end{aligned}$$

And from this point we finish the problem similarly to before.

#### **Note**

This was a lot of writing, but I hope you found a method in here that makes sense, and you understand that there are multiple ways to do the following problems. We will try to just show you the streamlined version of this in future examples, but feel free to try other substitutions if you want the practice!

The substitutions we will use (to avoid extra negatives) for this section will be

| Form                  | Trig Identity                         | Substitution                   | Assumptions                                                           | Triangle                                                                            |
|-----------------------|---------------------------------------|--------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $\sqrt{b^2x^2 - a^2}$ | $\sec^2(\theta) - 1 = \tan^2(\theta)$ | $x = \frac{a}{b} \sec(\theta)$ | $0 \leq \theta < \frac{\pi}{2},$<br>$\frac{\pi}{2} < \theta \leq \pi$ |  |
| $\sqrt{a^2 - b^2x^2}$ | $1 - \sin^2(\theta) = \cos^2(\theta)$ | $x = \frac{a}{b} \sin(\theta)$ | $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$                       |  |
| $\sqrt{a^2 + b^2x^2}$ | $\tan^2(\theta) + 1 = \sec^2(\theta)$ | $x = \frac{a}{b} \tan(\theta)$ | $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$                             |  |

Note: Most of the examples you will see in this class will have  $b = 1$ , but the above formulas are given for completion.

Let's do some examples.

### Example 7.3.1

Evaluate

$$\int \frac{x^3}{\sqrt{4 + x^2}} dx$$

### Solution

We see a weird root in the integral, so we know to try a trig sub. In this case, the form is  $\sqrt{4 + x^2} \rightsquigarrow \sqrt{a^2 + b^2x^2}$  where  $a = 2$  and  $b = 1$ . Then, in this form the substitution we use is

$$x = \frac{a}{b} \tan(\theta)$$

$$x = 2 \tan(\theta)$$

$$dx = 2 \sec^2(\theta) d\theta$$

This substitution gives us

$$\begin{aligned}
\int \frac{x^3}{\sqrt{4+x^2}} dx &= \int \frac{(2 \tan(\theta))^3}{\sqrt{4 + (2 \tan(\theta))^2}} 2 \sec^2(\theta) d\theta \\
&= 16 \int \frac{\tan^3(\theta)}{\sqrt{4(1 + \tan^2(\theta))}} \sec^2(\theta) d\theta \\
&= \frac{16}{2} \int \frac{\tan^3(\theta)}{\sqrt{\sec^2(\theta)}} \sec^2(\theta) d\theta \\
&= 8 \int \tan^3(\theta) \sec(\theta) d\theta, \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}
\end{aligned}$$

Now we have converted the integral into a trigonometric integral that we can solve using methods from Section 7.2. Now, we keep one factor of  $\tan(\theta) \sec(\theta)$  and rewrite everything else in terms of  $\sec(\theta)$ .

$$\begin{aligned}
\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int \tan^2(\theta) \tan(\theta) \sec(\theta) d\theta \\
&= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta
\end{aligned}$$

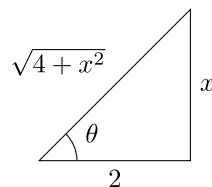
Now we have converted the trig integral into one we can solve with a  $u$ -sub where  $u = \sec(\theta)$  so  $du = \sec(\theta) \tan(\theta) d\theta$ . Thus,

$$\begin{aligned}
\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta \\
&= 8 \int u^2 - 1 du \\
&= 8 \left[ \frac{u^3}{3} - u \right] \\
&= 8 \left[ \frac{\sec^3(\theta)}{3} - \sec(\theta) \right]
\end{aligned}$$

Lastly, the question was in  $x$ , not  $\theta$ . So we need something to replace  $\sec(\theta)$  with. Looking back at our substitution, we have

$$x = 2 \tan(\theta) \Rightarrow \tan(\theta) = \frac{x}{2} = \frac{\text{opp}}{\text{adj}}$$

lets us build the triangle below.



So we can see that

$$\sec(\theta) = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{4+x^2}}{2}.$$

Finally the antiderivative is

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \left[ \frac{\sec^3(\theta)}{3} - \sec(\theta) \right] \\ &= 8 \left[ \frac{1}{3} \left( \frac{\sqrt{4+x^2}}{2} \right)^3 - \frac{\sqrt{4+x^2}}{2} \right] + C\end{aligned}$$

 **Note**

That was a really long problem. We can shorten some of the substitutions using the triangle at the end a bit sooner. Here is another way to solve the same problem:

 **Solution**

We see a weird root in the integral, so we know to try a trig sub. In this case, the form is  $\sqrt{4+x^2} \rightsquigarrow \sqrt{a^2+b^2x^2}$  where  $a=2$  and  $b=1$ . Then, in this form the substitution we use is

$$\begin{aligned}x &= \frac{a}{b} \tan(\theta) \\ x &= 2 \tan(\theta) \\ dx &= 2 \sec^2(\theta) d\theta\end{aligned}$$

Further, we have

$$x = 2 \tan(\theta) \Rightarrow \tan(\theta) = \frac{x}{2} = \frac{\text{opp}}{\text{adj}}$$

lets us build the triangle below Then, we can see that the square root term in our integral can be quickly replaced by a trig function:

$$\sec(\theta) = \frac{\sqrt{4+x^2}}{2} \Rightarrow \sqrt{4+x^2} = 2 \sec(\theta)$$

This substitution gives us

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= \int \frac{(2 \tan(\theta))^3}{2 \sec(\theta)} 2 \sec^2(\theta) d\theta \\ &= 8 \int \tan^3(\theta) \sec(\theta) d\theta, \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}\end{aligned}$$

Now we have converted the integral into a trigonometric integral that we can solve using methods from Section 7.2. Now, we keep one factor of  $\tan(\theta) \sec(\theta)$  and rewrite everything else in terms of  $\sec(\theta)$ .

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int \tan^2(\theta) \tan(\theta) \sec(\theta) d\theta \\ &= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta\end{aligned}$$

Now we have converted the trig integral into one we can solve with a  $u$ -sub where  $u = \sec(\theta)$  so  $du = \sec(\theta) \tan(\theta) d\theta$ . Thus,

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta \\ &= 8 \int u^2 - 1 du \\ &= 8 \left[ \frac{u^3}{3} - u \right] \\ &= 8 \left[ \frac{\sec^3(\theta)}{3} - \sec(\theta) \right]\end{aligned}$$

Lastly, the question was in  $x$ , not  $\theta$ . So we need something to replace  $\sec(\theta)$  with. Looking back at our triangle, we can see that

$$\sec(\theta) = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{4+x^2}}{2}.$$

Finally the antiderivative is

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \left[ \frac{\sec^3(\theta)}{3} - \sec(\theta) \right] \\ &= 8 \left[ \frac{1}{3} \left( \frac{\sqrt{4+x^2}}{2} \right)^3 - \frac{\sqrt{4+x^2}}{2} \right] + C\end{aligned}$$

### Note

A lot of the steps are the same, use whichever method above you prefer when solving these, but always draw the triangle as part of your work.

### Example 7.3.2

Evaluate

$$\int \frac{\sqrt{9-x^2}}{x^2} dx$$

### Solution

The form of the root in this problem is  $\sqrt{9-x^2} \rightsquigarrow \sqrt{a^2-b^2x^2}$ . So, we know that we need to use the substitution

$$\begin{aligned}x &= \frac{a}{b} \sin(\theta) \\ &= 3 \sin(\theta) \\ dx &= 3 \cos(\theta) d\theta\end{aligned}$$

The triangle is then

So, the square root part of the integral becomes  $\sqrt{9 - x^2} = 3 \cos(\theta)$ . The substitution gives us

$$\begin{aligned}\int \frac{\sqrt{9 - x^2}}{x^2} dx &= \int \frac{3 \cos(\theta)}{(3 \sin(\theta))^2} 3 \cos(\theta) d\theta \\ &= \int \cot^2(\theta) d\theta\end{aligned}$$

Now we can try solving this.

$$\begin{aligned}\int \cot^2(\theta) d\theta &= \int \csc^2(\theta) - 1 d\theta \\ &= \int \csc^2(\theta) d\theta - \int 1 d\theta \\ &= -\cot(\theta) - \theta\end{aligned}$$

Lastly, we need to get back to  $x$  for the final answer. Using the triangle from before we

7.3a Section Summary:

- We figured out which trigonometric substitutions would help us simplify problems without extra negatives.
- We used trig substitutions to simplify indefinite integrals.
- We used right triangles to rewrite answers in terms of the original variable.